Question 3 - CNN

Tevyn Vergara | 46421201 | Electrical & Biomedical Engineer

DISCLAIMER: Google Gemini 3 was used for this assignment for:

1. Layout tips (all questions)
2. Syntax checking (all questions)
3. Trouble shooting and debugging
4. Functionality tips and suggestions (all questions)
5. Tips and explanation for key functions/theories and purposes
6. Re-organizing to look more modular and readable

References:

1. Google. (2026, March 17). Methods of providing alterative text for accessibility on websites. [Generative AI chat]. Gemini 3. https://gemini.google.com/

Standard Imports for Assignment 2

Please run this before executing any other cell

In [18]:

```python
import numpy as np
import pickle
import matplotlib.pyplot as plt
import skimage
import sys
import time
import optuna
import sklearn
from sklearn.metrics import accuracy_score, confusion_matrix

print('numpy', np.__version__)
print('matplotlib', plt.matplotlib.__version__)
print('skimage', skimage.__version__)
print('sys', sys.version)
print('pickle', pickle.format_version)
print(time.ctime())
print('sklearn', sklearn.__version__)
print('optuna', optuna.__version__)
```

```
numpy 2.4.4
matplotlib 3.10.8
skimage 0.26.0
sys 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)]
pickle 4.0
Fri Apr 17 01:52:59 2026
sklearn 1.8.0
optuna 4.8.0
```

NOTE: Code was ran locally as I have a dedicated GPU

Question "0" - CIFAR-10 Dataset

Here, I'm attempting to load the images. The formatting is currently:

- Labels in batches.meta [0-9] with their respective image classes: 0: Aeroplane 1: Automobile 2: Bird 3: Cat 4: Deer 5: Dog 6: Frog 7: Horse 8: Ship 9: Truck
- 3072 features of flattened information --> 32 x 32 pixel images with 3 RGB channels
- 50,000 training images split into x5 data_batch_x files (5 in total). Format: shape(10,000, 3072). So 10,000 images per batch of 3072 flattened information

To format for CNN training:

1. Use unpickle() from source: https://www.cs.toronto.edu/~kriz/cifar.html. This returns a batch dictionary of the 10,000 images, their labels, and each of their single channel information of (32x32 | 3 RGB). Their is also metadata for the batch such as the image and batch file names (Not needed for project)
2. Unload image class labels from batches.meta (0-9) and convert their byte strings to normal text for readibility. Store in meta.
3. Unload and also combine the 5 batches of training data from data_batch_x where x: [0,5]. Assuming that all the images have the same format of (32x32 |3 RGB), append all imagee (with their data) and their respective labels to a dictionary of arrays that contain:
    - Image data (batch[b"data"])
    - Image labels (batch[b"data"])
    - Metadata (filenames, image names) but unused

To ensure that testing data was loaded correctly, loop through the whole "batch" dataset and load the first image of each class. Display it and its respective label.

In [21]:

```python
import os
import pickle
import numpy as np
import matplotlib.pyplot as plt


# Utility: read one CIFAR-10 batch/metadata pickle file from disk. From: https://www.cs.toronto.edu/~kriz/cifar.html
def unpickle(file_path):
    with open(file_path, "rb") as fo:
        batch = pickle.load(fo, encoding="bytes")
    return batch

# End-to-end loader for CIFAR-10 in the provided folder structure.
# Returns train/test images + labels, and human-readable class names.
def load_cifar10(data):
    # 1) Load class label names (airplane, automobile, etc.) from metadata.
    meta = unpickle(os.path.join(data, "batches.meta"))
    label_names = [name.decode("utf-8") for name in meta[b"label_names"]]

    # 2) Read and concatenate all 5 training batch files.
    train_images = []
    train_labels = []
    for i in range(1, 6):
        batch = unpickle(os.path.join(data, f"data_batch_{i}"))
        train_images.append(batch[b"data"])
        train_labels.extend(batch[b"labels"])

    # 3) Stack image rows into one array of shape (50000, 3072).
    X_train = np.vstack(train_images)
    y_train = np.array(train_labels, dtype=np.int64)

    # 3) Load test batch
    test_batch = unpickle(os.path.join(data, "test_batch"))
    X_test = test_batch[b"data"]
    y_test = np.array(test_batch[b"labels"], dtype=np.int64)

    return X_train, y_train, X_test, y_test, label_names

# Load the CIFAR-10 Dataset
data = "cifar-10-batches-py"
X_train, y_train, X_test, y_test, label_names = load_cifar10(data)

# Quick shape sanity check to confirm successful parsing.
print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_test shape: {y_test.shape}")
print("Label names:", label_names)

# Visual inspection: show one example image from each class.
fig, axes = plt.subplots(2, 5, figsize=(14, 6))
for class_id, ax in enumerate(axes.ravel()):
    # Pick the first training sample for this class index.
    idx = np.where(y_train == class_id)[0][0]
    img = X_train[idx].reshape(3, 32, 32).transpose(1, 2, 0)  # (3072,) -> (32,32,3)
    ax.imshow(img)
    ax.set_title(f"{class_id}: {label_names[class_id]}")
    ax.axis("off")

plt.tight_layout()
plt.show()
```

```
C:\Users\tevyn\AppData\Local\Temp\ipykernel_22504\1026889439.py:10: VisibleDeprecationWarning: dtype(): align should be passed as Python or NumPy boolean but got `align=0`. Did you mean to pass a tuple to create a su
barray type? (Deprecated NumPy 2.4)
  batch = pickle.load(fo, encoding="bytes")
X_train shape: (50000, 3072)
y_train shape: (50000,)
X_test shape: (10000, 3072)
y_test shape: (10000,)
Label names: ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
```

0: airplane
1: automobile
2: bird
3: cat
4: deer
5: dog
6: frog
7: horse
8: ship
9: truck

Here is my application of a NNN.

The simple 3-layer structure is:

1. Input layer of 3072 neurons (veectorized image of 32 x 32 x 3RGB image data)
2. Hidden layer of fully connected neurons. Size tuned by Optuna and activated with ReLU activation function
3. Output layer - 10 neurons (one per CIFAR-10 class) and activated by Softmax

Weights are initialized using **He initialization** — scaled by $\sqrt{2/n}$ — which is specifically designed for ReLU networks to keep activations from collapsing or exploding early in training. It prevents vanishing/exploding gradients by setting weight variance, creating a Gaussian distribution with zero mean

Forward Pass

The network computes two linear transforms separated by a non-linearity:

Softmax converts raw scores into a probability distribution across the 10 classes:

Loss Function

The loss combines **Cross-Entropy** with **L2 Regularization** a penalty term, equal to the sum of squared weights which forces weight parameters towards zero. This helps in noise robustness for the model.

Backpropagation

Gradients flow backwards through the network using the chain rule:

1. **Output layer gradient** — the softmax + cross-entropy combo has a clean derivative: $dZ_2 = A_2 - Y_{\text{one-hot}}$
2. **Weight gradients** with L2 regularization
3. **Hidden layer gradient** — ReLU's derivative is just a mask: 1 where $Z_1 > 0$, else 0

Weights are then updated via standard **Gradient Descent**

Hyperparameter Tuning with Optuna

Rather than guessing hidden layer size and learning rate, **Optuna** runs 10 trials on a 5,000-image subset. Each trial trains for 20 epochs and evaluates test accuracy. The best configuration is then used for the full training run.

Evaluation

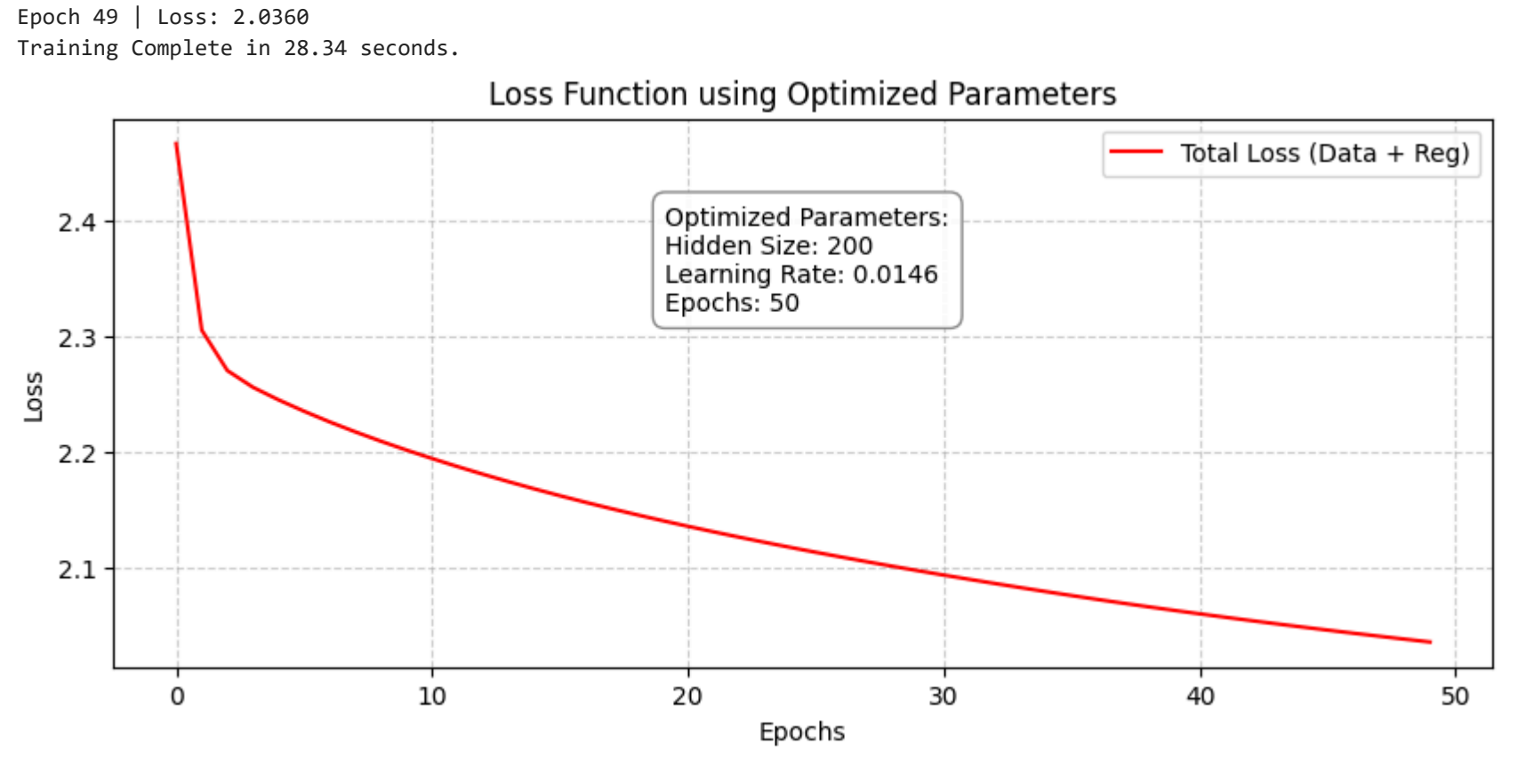
After training, predictions are generated with a final forward pass. Results are shown in two plots:

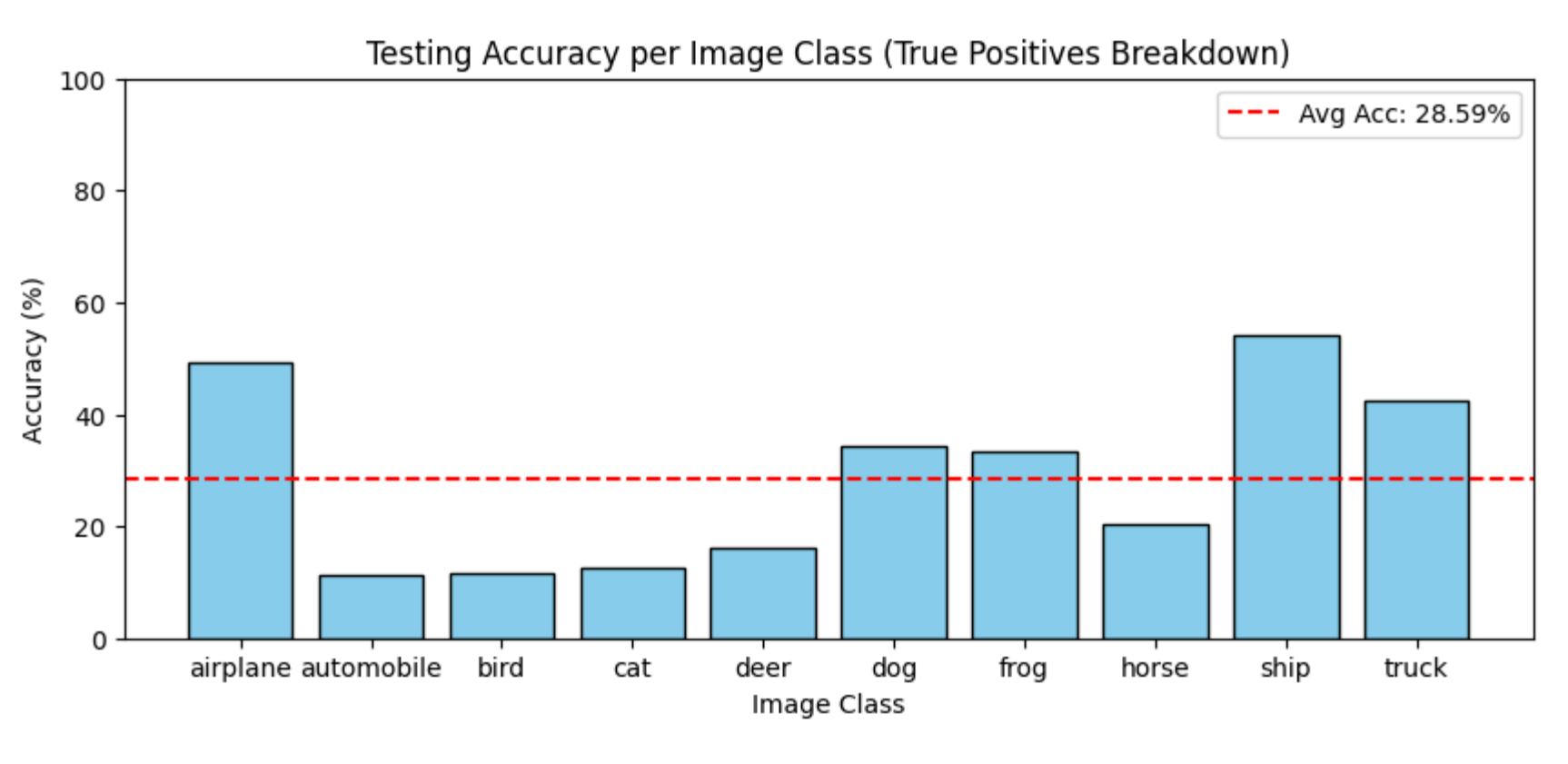
- Loss curve over epochs
- Per-class accuracy breakdown (true positives for each of the 10 CIFAR-10 classes)

In [ ]:

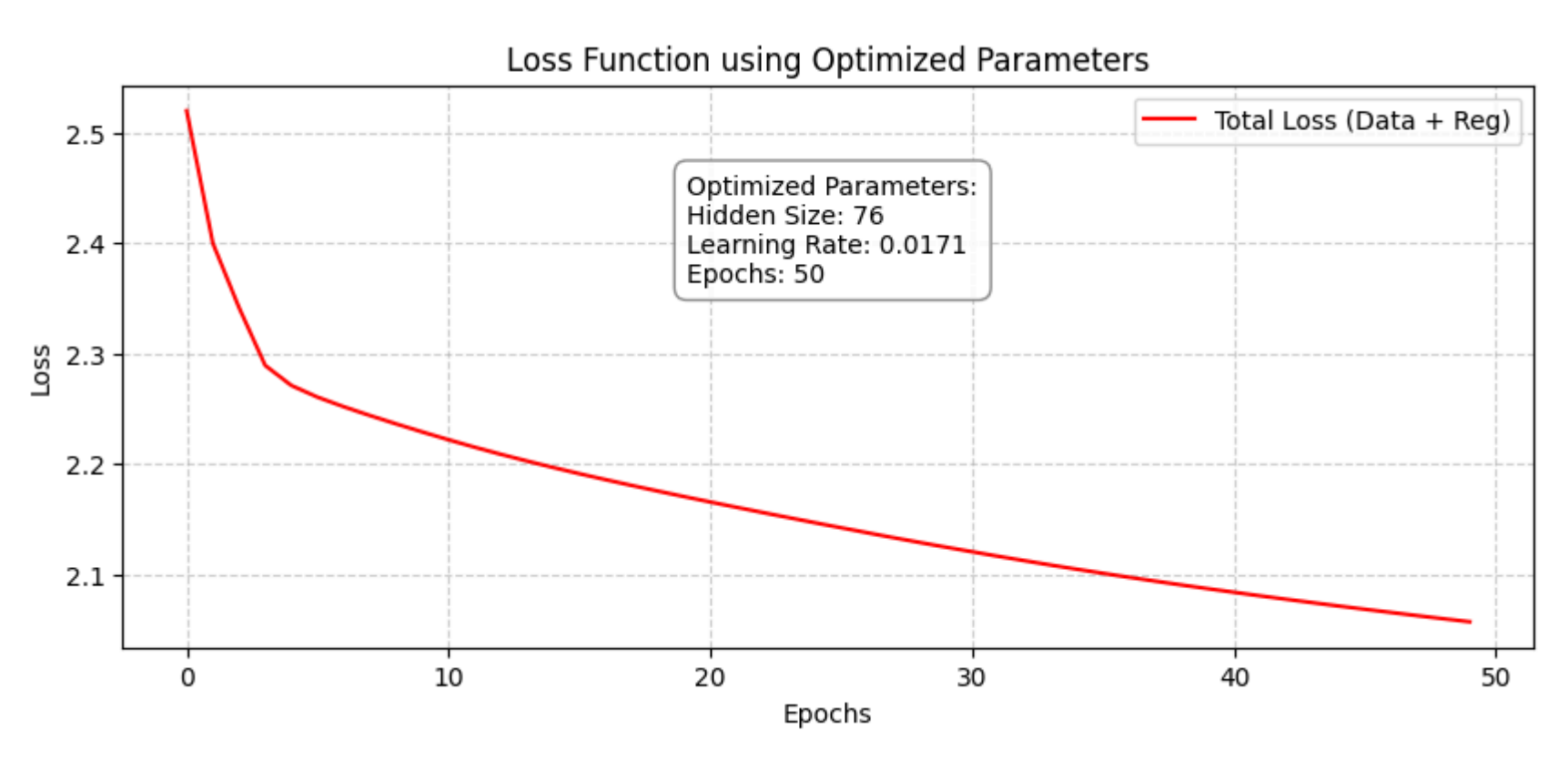
```python
################## Global Hyperparameters ##################
EPOCHS = 50  # Number of training epochs for the main training loop (after Optuna optimization)

################## Activation Functions ##################
def relu(Z):
    """Non-linear transform map activation function (ReLU). Drops negative activations to 0."""
    return np.maximum(0, Z)

def relu_backward(dA, Z):
    """Backpropagation gradient for ReLU (derivative is 1 for Z>0, else 0)."""
    dZ = np.array(dA, copy=True)
    dZ[Z <= 0] = 0
    return dZ

def softmax(Z):
    """Purpose: Neural linear classifier for output layer."""
    expZ = np.exp(Z - np.max(Z, axis=1, keepdims=True)) # Shift for numerical stability
    return expZ / np.sum(expZ, axis=1, keepdims=True)

################## Neural Network Class ##################
class NeuralNetwork:
    """
    3-Layer Architecture:
    1. Input Layer (Implicit 3072 neurons from flattened CIFAR-10)
    2. Hidden Layer (W1, b1)
    3. Output Layer (W2, b2)
    """
    def __init__(self, input_size=3072, hidden_size=128, output_size=10, lambda_reg=0.01):
        """Initializes Learning Biases and W1/W2 Weights (He Initialization for ReLU)"""
        self.W1 = np.random.randn(input_size, hidden_size) * np.sqrt(2. / input_size)
        self.b1 = np.zeros((1, hidden_size))
        self.W2 = np.random.randn(hidden_size, output_size) * np.sqrt(2. / hidden_size)
        self.b2 = np.zeros((1, output_size))
        self.lambda_reg = lambda_reg  # L2 Regularization parameter to prevent overfitting

    def forward(self, X):
        """
        Forward propagation through the network mapping features to predictions
        """
        # Hidden Layer Transform
        self.Z1 = np.dot(X, self.W1) + self.b1
        self.A1 = relu(self.Z1)

        # Output Layer Neural Classifier
        self.Z2 = np.dot(self.A1, self.W2) + self.b2
        self.A2 = softmax(self.Z2)
        return self.A2

    def compute_total_loss(self, Y_hat, Y_true):
        """
        Calculates Total Loss: Cross-Entropy Loss + Data Regularization
        """
        m = Y_true.shape[0]
        # Cross entropy data loss
        log_probs = -np.log(Y_hat[range(m), Y_true] + 1e-8)
        data_loss = np.sum(log_probs) / m

        # Penalize large weights to avoid overfitting
        reg_loss = (self.lambda_reg / (2 * m)) * (np.sum(np.square(self.W1)) + np.sum(np.square(self.W2)))
        return data_loss + reg_loss

    def backward(self, X, Y_true, learning_rate):
        """
        Computes Back propagation to learn W1 and W2 gradients
        Also update weights with gradient descent
        """
        m = X.shape[0]

        # 1. Output Layer Gradients
        dZ2 = self.A2.copy()
        dZ2[range(m), Y_true] -= 1
        dZ2 /= m

        dW2 = np.dot(self.A1.T, dZ2) + (self.lambda_reg / m) * self.W2
        db2 = np.sum(dZ2, axis=0, keepdims=True)

        # 2. Hidden Layer Gradients
        dA1 = np.dot(dZ2, self.W2.T)
        dZ1 = relu_backward(dA1, self.Z1)

        dW1 = np.dot(X.T, dZ1) + (self.lambda_reg / m) * self.W1
        db1 = np.sum(dZ1, axis=0, keepdims=True)

        # 3. Update parameters using standard Gradient Descent
        self.W1 -= learning_rate * dW1
        self.b1 -= learning_rate * db1
        self.W2 -= learning_rate * dW2
        self.b2 -= learning_rate * db2

########################## Training, Optimization and Evaluation ###########################
def run_optimization_and_training(X_train_raw, y_train, X_test_raw, y_test, label_names, epochs=EPOCHS):
    """
    Normalizes data, runs Optuna for hyperparameter tuning, trains the NN, and plots results.
    """
    # Normalization: Scale pixel values to [0, 1] for stable gradients
    print("Normalizing Data...")
    X_train_norm = X_train_raw / 255.0
    X_test_norm = X_test_raw / 255.0

    # 1. OPTUNA Hyperparameter Optimization
    def objective(trial):
        """
        Optuna objective function to find best hidden_size and learning_rate
        """
        hidden_size = trial.suggest_int("hidden_size", 64, 256)
        learning_rate = trial.suggest_float("learning_rate", 1e-3, 1e-1, log=True)

        nn = NeuralNetwork(hidden_size=hidden_size)
        # Train on a small 5,000 image subset for rapid optuna trials
        idx = np.random.choice(X_train_norm.shape[0], 5000, replace=False)
        X_sub, y_sub = X_train_norm[idx], y_train[idx]

        for _ in range(20):
            preds = nn.forward(X_sub)
            nn.backward(X_sub, y_sub, learning_rate)

        preds_test = nn.forward(X_test_norm)
        acc = accuracy_score(y_test, np.argmax(preds_test, axis=1))
        return acc

    # Run Optuna optimization to find best hyperparameters
    print("Running Optuna Optimization...")
    optuna.logging.set_verbosity(optuna.logging.WARNING) # Reduce output clutter
    study = optuna.create_study(direction="maximize")
    study.optimize(objective, n_trials=10)
    best_params = study.best_params
    print("Best Parameters:", best_params)

    # Main training Loop with optimized hyperparameters from above
    nn = NeuralNetwork(hidden_size=best_params['hidden_size'])
    loss_history = []

    print("Starting Main Training Loop...")
    start_time = time.time()
    for epoch in range(epochs):
        preds = nn.forward(X_train_norm)
        loss = nn.compute_total_loss(preds, y_train)
        loss_history.append(loss)

        # Learn gradients and update weights
        nn.backward(X_train_norm, y_train, best_params['learning_rate'])

        if epoch % 10 == 0 or epoch == epochs - 1:
            print(f"Epoch {epoch} | Loss: {loss:.4f}")

    print(f"Training Complete in {time.time() - start_time:.2f} seconds.")

    # Evaluate on test set with final trained model
    test_preds = nn.forward(X_test_norm)
    test_predictions_classes = np.argmax(test_preds, axis=1)

    # Loss Function with Optuna parameters displayed
    plt.figure(figsize=(10, 4))
    plt.plot(loss_history, label='Total Loss (Data + Reg)', color='red')
    plt.title('Loss Function using Optimized Parameters')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')

    # Textbox displaying the hyperparameters
    param_text = f"Optimized Parameters:\nHidden Size: {best_params['hidden_size']}\nLearning Rate: {best_params['learning_rate']:.4f}\nEpochs: {epochs}"
    plt.text(0.40, 0.65, param_text, transform=plt.gca().transAxes, fontsize=10,
             bbox=dict(facecolor='white', alpha=0.9, edgecolor='gray', boxstyle='round,pad=0.5'))

    plt.grid(True, linestyle='--', alpha=0.6)
    plt.legend(loc='upper right')
    plt.show()

    # Accuracies as true positives per class with average accuracy of model
    avg_acc = accuracy_score(y_test, test_predictions_classes)
    conf_matrix = confusion_matrix(y_test, test_predictions_classes)

    # Convert accuracies to percentages
    class_accuracies_percent = (conf_matrix.diagonal() / conf_matrix.sum(axis=1)) * 100

    plt.figure(figsize=(10, 4))
    plt.bar(label_names, class_accuracies_percent, color='skyblue', edgecolor='black')
    plt.axhline(y=avg_acc * 100, color='r', linestyle='--', label=f'Avg Acc: {avg_acc*100:.2f}%')
    plt.title('Testing Accuracy per Image Class (True Positives Breakdown)')
    plt.xlabel('Image Class')
    plt.ylabel('Accuracy (%)')
    plt.ylim(0, 100) # Set limits correctly for percentages
    plt.legend()
    plt.show()

# ==========================================
# EXECUTE (Assuming X_train, y_train, X_test, y_test, label_names are loaded)
# ==========================================
run_optimization_and_training(X_train, y_train, X_test, y_test, label_names, epochs=EPOCHS)
```

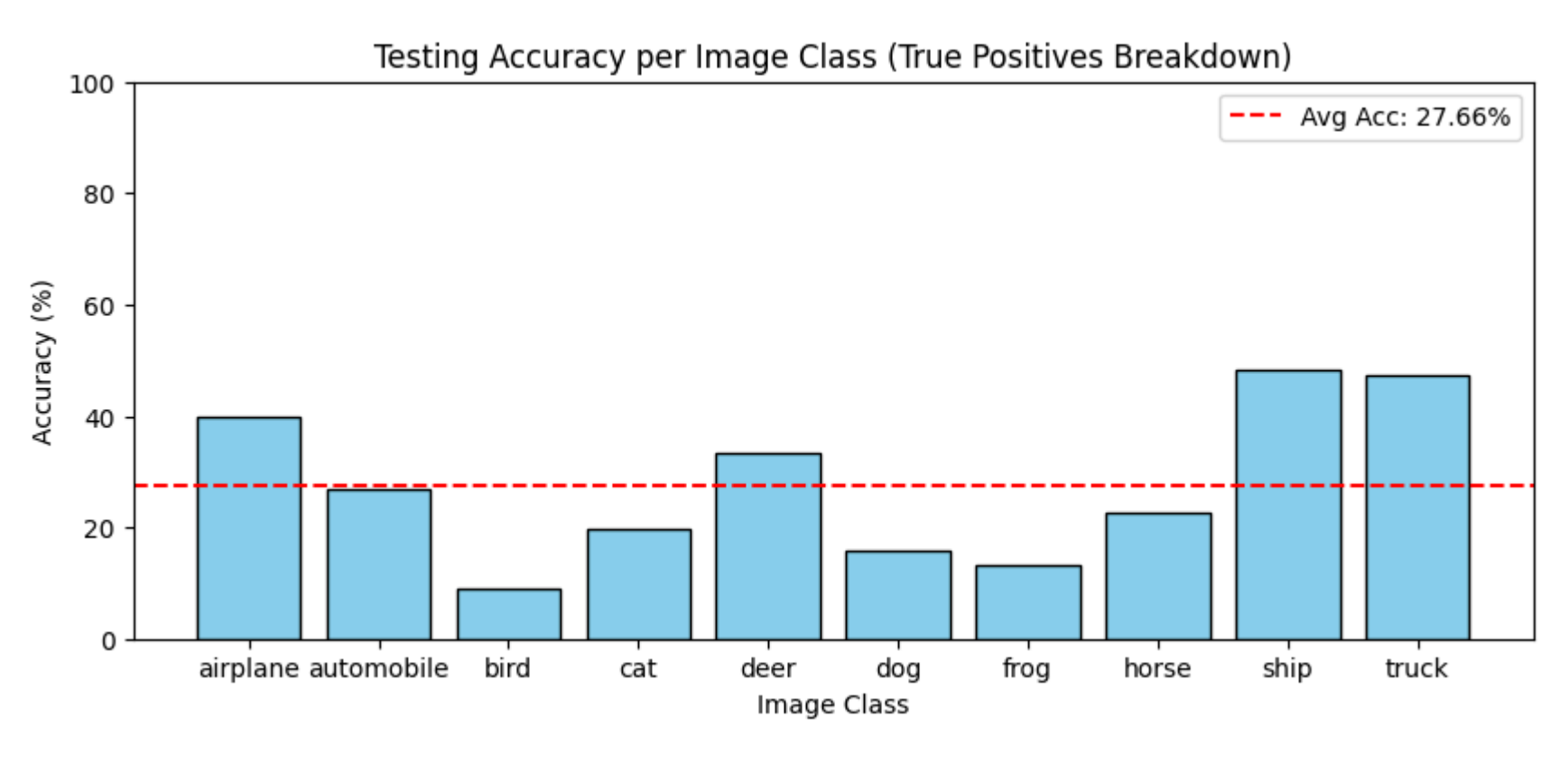
```
Normalizing Data...
Running Optuna Optimization...
Best Parameters: {'hidden_size': 200, 'learning_rate': 0.014621507294916212}
Starting Main Training Loop...
Epoch 0 | Loss: 2.4658
Epoch 10 | Loss: 2.1942
Epoch 20 | Loss: 2.1358
Epoch 30 | Loss: 2.0937
Epoch 40 | Loss: 2.0603
Epoch 49 | Loss: 2.0360
Training Complete in 28.34 seconds.
```



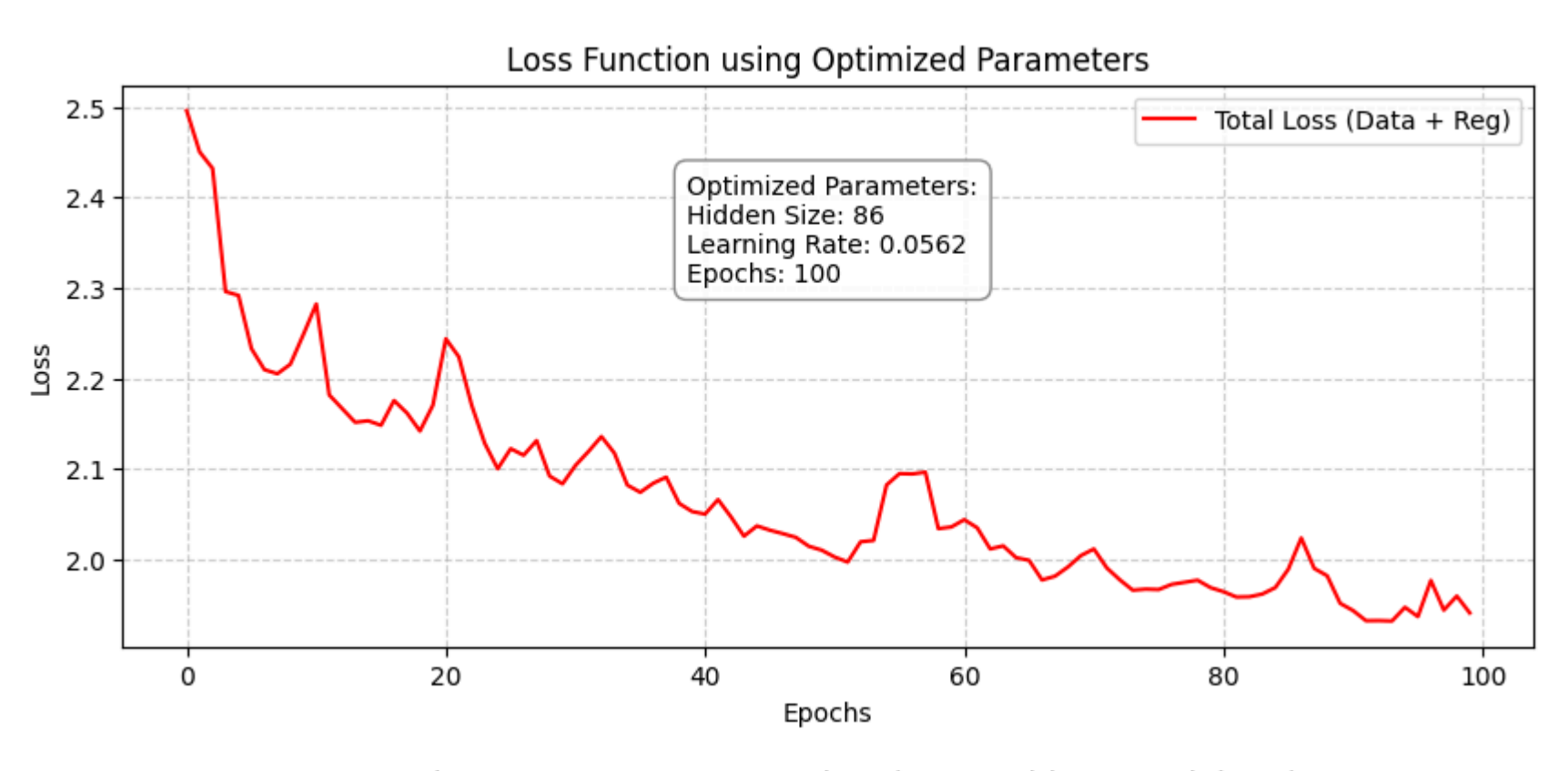


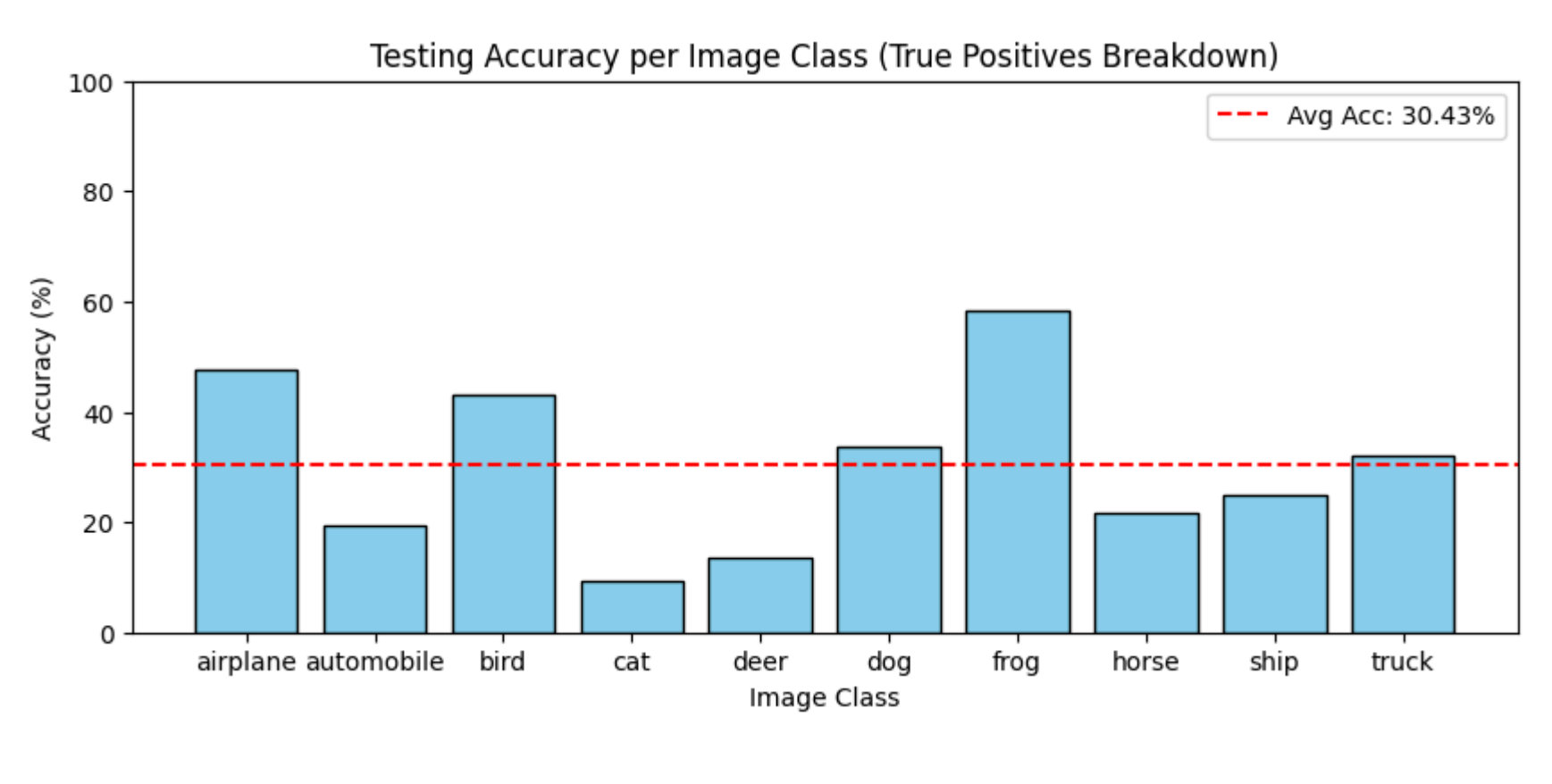
EPOCH: 50



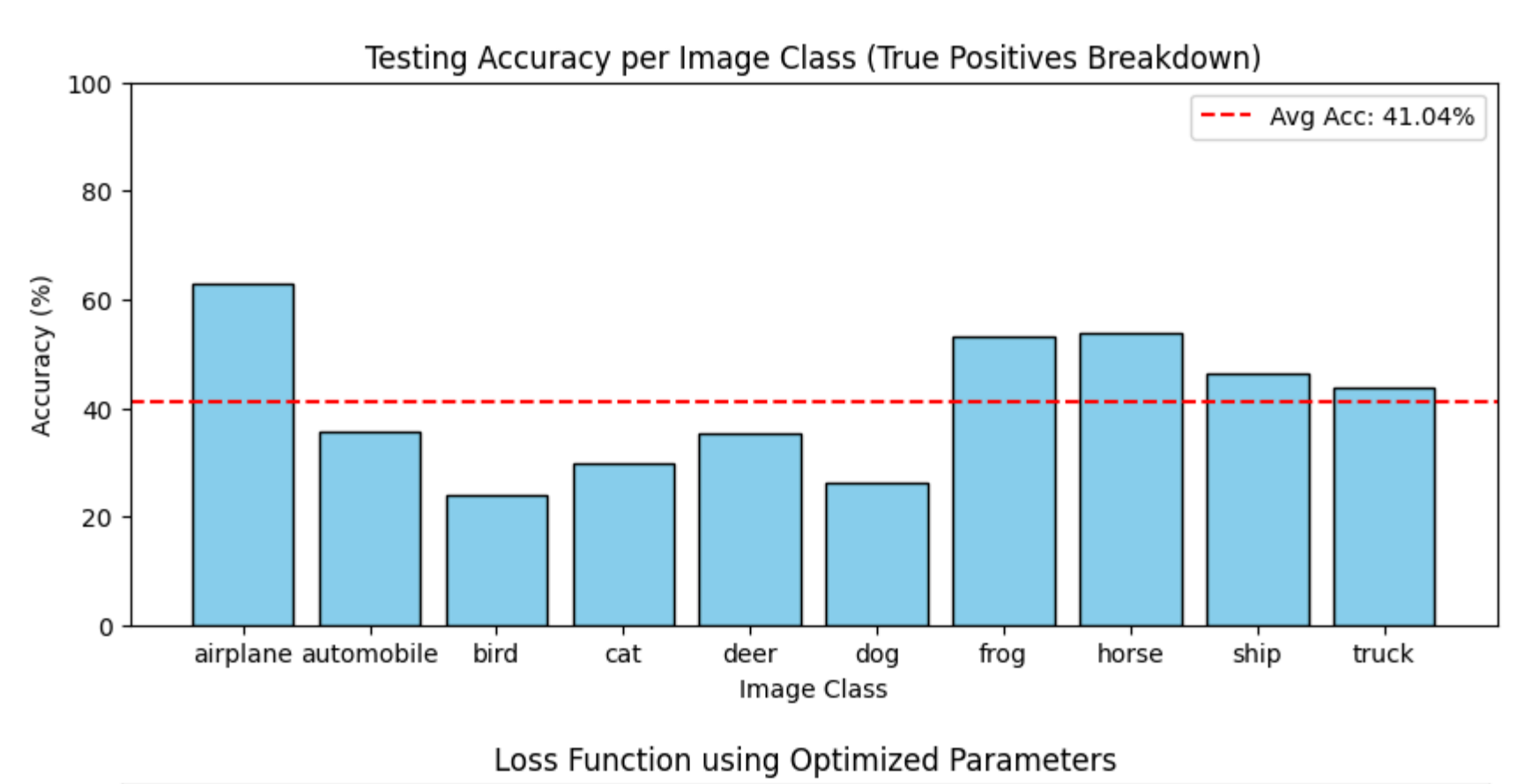


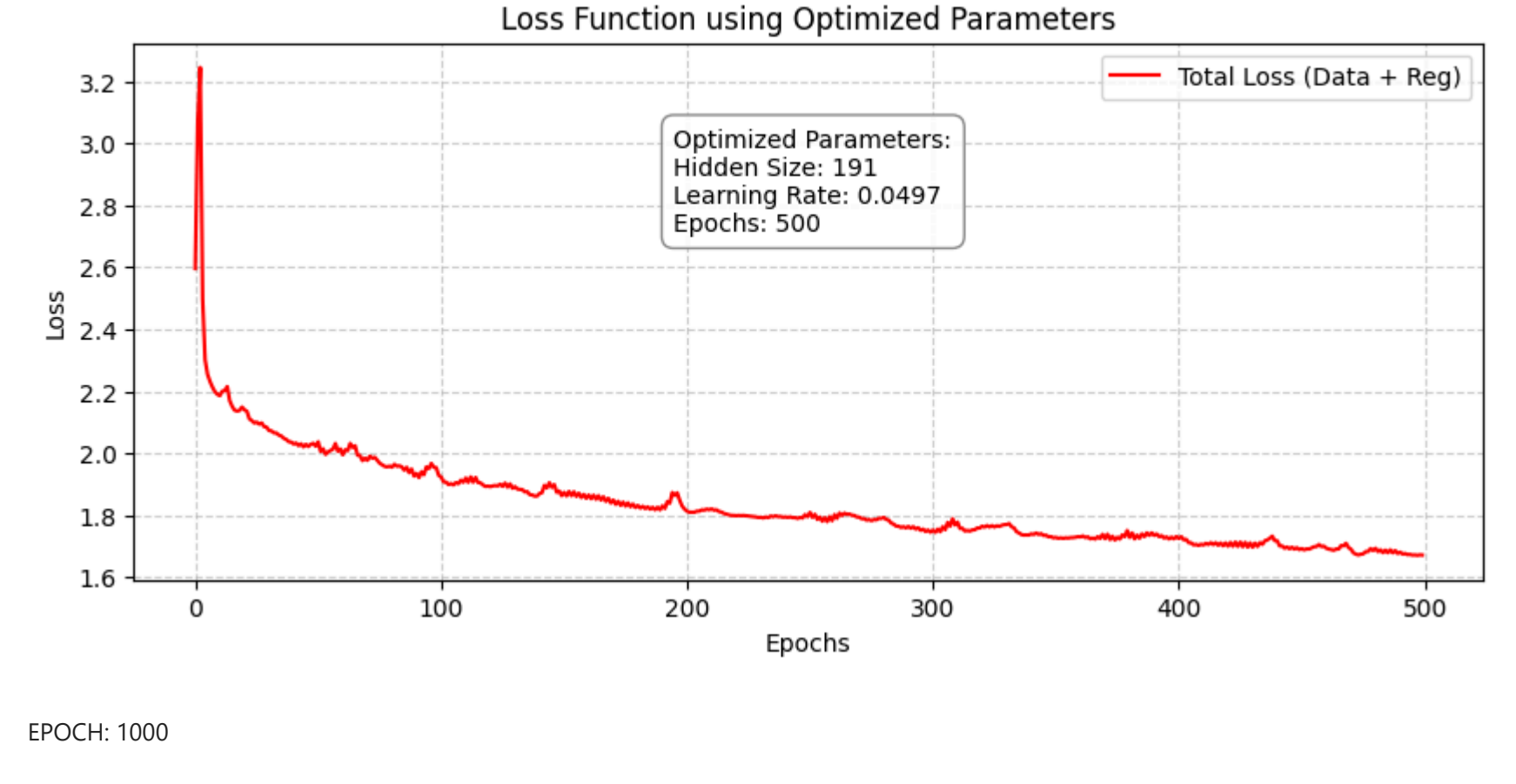
EPOCH: 100



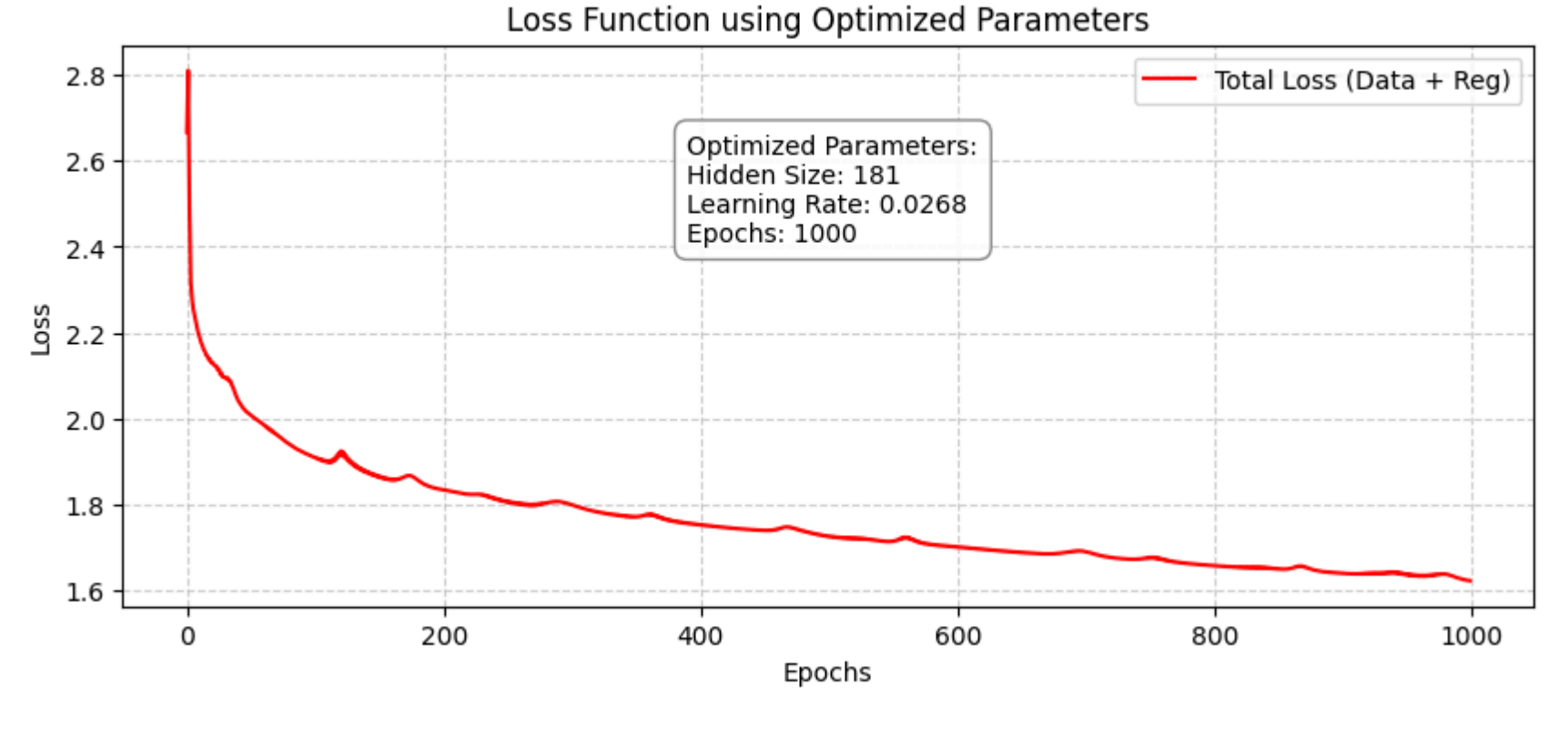


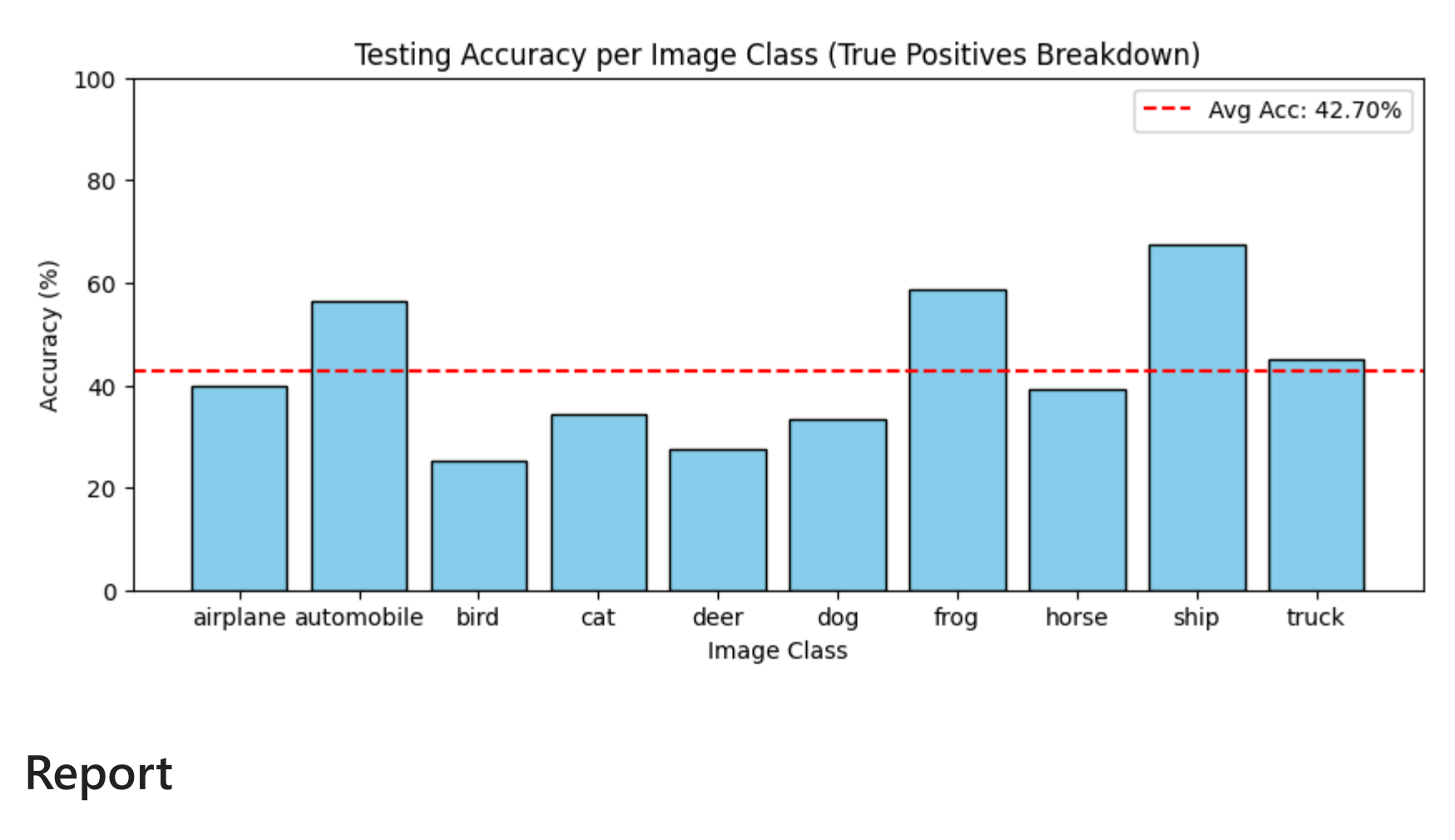
EPOCH: 500





EPOCH: 1000





Report

1. Design Choices

If I wanted to tweak the architecture further, Leaky ReLU or GELU could have potentially been added and implemented however ReLU was chosen for its popularity (and tutor recommendation).

2. Training Results & Limitations

The network was tested with 50, 100, 500, and 1000 epochs. Looking at the graphs, the testing accuracy clearly hits a bottleneck around the 500-epoch mark at around 43% Plateaus like this—often hovering near 50% on CIFAR-10—aren't really a hyperparameter tuning issue; it's an architectural limit. The network lacks the depth to capture highly complex spatial features in the images.

This task restricted the NN to just 1 hidden layer. This forced the model to run sequentially rather than building a network of hundreds of layers with skip connections and pooling mechanisms. It didn't fully allow the model to develop comprehensive recognition of edges -> textures --> recognizable objects DL frameworks (such as PyTorch and Tensor) are able to achieve higher accuracy and save computation time since they can handle massive, parallel matrix operations and complex calculus automatically. These would have allowed the network to scale up the capacity needed to acheive extremely high testing accuracy